

MATH 4790: Quiz 1

January 23, 2026

Name:

Score: _____/15

Instructions: Show all work and justify your answers. Only calculators approved for Exam FM are permitted.

Question 1.

A three-year bond pays annual coupons of \$50 and is redeemable at \$1000 at maturity. The bond yields an effective annual interest rate of 8%.

Find the modified and Macaulay duration of the bond. *Round your final answer to two decimals.*

Question 2.

The following table gives the annual *effective* spot rates s_n for maturities $n = 1, 2, \dots, 6$.

n	1	2	3	4	5	6
s_n	4.0%	4.2%	4.4%	4.6%	4.8%	5.0%

A set of cash flows consists of \$100 at time $t = 1$, \$200 at time $t = 2$, and \$500 at time $t = 5$.

Determine the value of this cash-flow stream at time $t = 3$. *Round your final answer to two decimals.*

Question 3.

An investor has liabilities of \$2,000 due at time $t = 1$, \$2,000 due at time $t = 2$, and \$4,000 due at time $t = 3$.

To exactly match these liabilities, the investor considers purchasing the following three bonds:

- (i) A one-year par value bond with a coupon rate of 3% and an effective annual interest rate of 6%.
- (ii) A two-year zero-coupon bond with an effective annual interest rate of 7%.
- (iii) A three-year par value bond with a coupon rate of 4% and an effective annual interest rate of 8%.

Calculate the total purchase price of the three bonds required to exactly match the liabilities.
Round your final answer to two decimals.